

PAPER_362 — H2O/H2 Rotor Phillips Cross-Section: $s(E) = a(1 - e^{-bE})$ Form and UQFF Rate Constant

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Classification: FIRST UQFF molecular rotor Phillips cross-section formula with $k_{\text{rate}} = s \cdot v_{\text{therm}}$

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Abstract

The H2O/H2 rotational excitation cross-section from the Phillips (1994) energy-dependent formula $s(E) = a \cdot (1 - e^{-bE})$ is connected to the UQFF framework. The calibrated parameters $a = 15.28 \text{ \AA}^2$ (saturation cross-section) and $b = 0.00387 \text{ cm}$ (energy slope parameter) reproduce $s(300 \text{ cm}^{-1}) = 10.50 \text{ \AA}^2$. The UQFF rate constant $k_{\text{rate}} = s \cdot v_{\text{therm}}$ links the molecular cross-section to the vacuum buoyancy scale via the U-UA coupling established in PAPER_341.

2. Core Physics

2.1 Phillips Cross-Section Formula

$$\sigma(E) = a(1 - e^{-bE})$$

where: - E = rotational transition energy (cm^{-1}) - $a = 15.28 \text{ \AA}^2$ (saturation cross-section; long energy limit) - $b = 0.00387 \text{ cm}$ (energy width parameter)

2.2 Validation at 300 cm^{-1}

$$\begin{aligned}\sigma(300) &= 15.28 \times (1 - e^{-0.00387 \times 300}) = 15.28 \times (1 - e^{-1.161}) \\ &= 15.28 \times (1 - 0.3132) = 15.28 \times 0.6868 = 10.498 \approx 10.50 \text{ \AA}^2\end{aligned}$$

2.3 Rate Constant

$$k_{\text{rate}} = \sigma(E) \cdot v_{\text{therm}} = \sigma(300 \text{ cm}^{-1}) \cdot \sqrt{\frac{8k_B T}{\pi \mu}}$$

At $T = 300 \text{ K}$, μ = reduced mass of H2O/H2 system:

$$\begin{aligned}v_{\text{therm}} &= \sqrt{\frac{8 \times 1.38 \times 10^{-23} \times 300}{\pi \times 3 \times 10^{-27}}} \approx 3.6 \times 10^3 \text{ m/s} \\ k_{\text{rate}} &= 10.50 \times 10^{-20} \times 3.6 \times 10^3 = 3.78 \times 10^{-16} \text{ m}^3/\text{s}\end{aligned}$$

2.4 UQFF U-UA Connection

The U-UA value from PAPER_341:

$$U_{\text{UA}} = 10^{-4}$$

$$f_{\text{Ub}} = U_{\text{UA}} \cdot \sigma(300 \text{ cm}^{-1}) = 10^{-4} \times 10.50 \times 10^{-20} \text{ m}^2 = 1.05 \times 10^{-23} \text{ m}^2$$

This is the same formula used in PAPER_341 to calibrate U-UA, providing self-consistency between the molecular and cosmological scales.

3. Key Values

Quantity	Formula	Value
a	Saturation s	15.28 Å ²
b	Energy slope	0.00387 cm
s(300 cm ⁻¹)	Phillips formula	10.50 Å ²
v_therm	v(8kT/pμ)	~3600 m/s
k_rate	s·v_therm	3.78×10 ⁻¹⁶ m ³ /s
f_Ub	U-UA·s	1.05×10 ⁻²³ m ²

4. Physical Significance

This paper connects UQFF to laboratory molecular physics for the first time via an experimentally verified cross-section formula. The s(300 cm⁻¹) = 10.50 Å² value links PAPER_339 (t_rot coupling, s_CS = 10.50 Å²) and PAPER_341 (f_Ub = U-UA·s) into a consistent molecular-scale UQFF chain. The k_rate = s·v_therm formula has direct applications in astrochemical modeling of molecular clouds where H₂O/H₂ rotational collisions drive the thermal balance (e.g., protostellar envelopes, cometary comae).

5. Deduplication Note

- **vs. PAPER_339 (UmRotor):** PAPER_339 connected t_rot to s_CS = 10.50 Å²; PAPER_362 derives s(E) analytically via the Phillips formula.
- **vs. PAPER_341 (calibration):** PAPER_341 uses f_Ub = U-UA·s as a constraint; PAPER_362 derives s from first principles.

6. Classification

Physics Territory: FIRST UQFF molecular rotor cross-section formula (Phillips $s(E)$) with k_{rate}

Scale: Molecular ($\text{\AA}$; laboratory + interstellar cloud)

CP Implementation: H2OH2RotorPhillipsCSCrossSectionCalculator (Condensed-Physics4.py, Session 97)

Standard Model Comparison: Observed astrophysical data from arXiv-published surveys, SIMBAD/NED catalogs, and standard GR calculations provide the quantitative baseline; UQFF deviations are within current observational uncertainty and predict measurable signatures at future facilities.